

Summary:

Patients who underwent ACLR with allograft compared to autograft had a 2-times higher risk of undergoing a TKA during follow-up, and additional risk factors for TKA after ACLR included patient age, female gender, concomitant chondral injury, and ipsilateral reoperation or contralateral surgery during follow-up.

Data:

ABSTRACT Background: Patients who undergo anterior cruciate ligament reconstruction (ACLR) are at a higher risk of undergoing total knee arthroplasty (TKA) and are at risk earlier than those from a general population. As ACLR patients become older, and as ACLR becomes more prevalent in the older athlete, the rates of TKA after ACLR will only increase. We aim to determine the incidence of TKA following ACLR by patient age at the time of ACLR and graft selection, as well as evaluate the risk factors for TKA after ACLR. **Methods:** Data from a US healthcare system's ACLR registry was used to conduct a cohort study. Primary ACLR patients were identified (2005-2022). Patient factors considered included age, body mass index (BMI), gender, race/ethnicity, smoking status, ASA classification, activity at the time of injury, and medical comorbidities. Time from injury to ACLR, concomitant meniscal or chondral injuries, multi-ligament injury, graft type, and drilling technique were procedure factors evaluated. Postoperative factors included revision surgery, ipsilateral reoperation, and contralateral operation during follow-up. The outcome of interest was a subsequent TKA. Patients were followed until the outcome of interest unless censored for membership disenrollment, death, or December 31, 2022 (the study end date). Multivariable Cox proportional hazards regression was used to determine factors associated with TKA following ACLR using $p < 0.05$ as the threshold for statistical significance. **Results:** The study sample included 52,222 primary ACLR. The mean age was 28.6 years, and more patients were male (60.2%). When considering age at the time of ACLR, the 15-year cumulative incidence of TKA after ACLR was 0.07% for those aged <30 years, 1.17% for those aged 30-39 years, 3.96% for those aged 40-49 years, 8.05% for those aged 50-59 years, and 22.53% for those aged ≥ 60 years. When considering graft used at ACLR, the 15-year cumulative incidence of TKA after ACLR was 3.15% for allograft, 0.36% for BPTB autograft, and 0.69% for hamstring autograft; it was 0.55% at 10-years follow-up for hybrid grafts (using an allograft and hamstring autograft) and 0.27% at 4-years for quadriceps tendon autograft. Risk factors for TKA included increasing age compared to those <40 years (40-49 years: hazard ratio [HR]=7.85, 95% confidence interval [CI]=4.65-13.24; 50-59 years: HR=17.51, 95% CI=10.47-29.28; ≥ 60 years: HR=50.41, 95% CI=25.52-99.60), female gender (HR=1.48, 95% CI=1.10-1.97), a history of other neurological disorders at the time of ACLR (HR=5.98, 95% CI=2.56-13.98), chondral injuries reported during the ACLR (HR=1.61, 95% CI=1.17-2.23), and allograft selection (HR=2.24, 95% CI=1.19-4.20). Ipsilateral reoperation (HR=2.46, 95% CI=1.80-3.35) and contralateral surgery (HR=2.55, 95% CI=1.80-3.61) during follow-up were both risk factors for TKA. **Conclusions:** The 15-year cumulative risk of TKA in patients who were ≥ 60 -years old when undergoing their ACLR was 23%. Patients who underwent ACLR with allograft had a 2-times higher risk of undergoing a TKA. Age, female gender, concomitant chondral injury, and ipsilateral reoperation or contralateral surgery during follow-up were additional risk factors for TKA in patients who have undergone prior ACLR.

Category: Knee - ACL**Outcomes from the DOSTAR Study: A Double Blinded Randomised Controlled Trial Comparing Dual and Single Tendon Hamstring Graft Anterior Cruciate Ligament Reconstruction**

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Disclosure information for authors is available at <https://isakos.com/2025/Papers>.

Summary:

Single Tendon Hamstring Grafts for ACL Reconstruction using Tibial Adjustable Button Fixation results in Significantly Lower Donor Site Morbidity

Data:

Introduction: In spite of the popularity of Hamstring Tendon Grafts for Anterior Cruciate Ligament (ACL) reconstruction, hamstring tendon harvest has been linked to reduced strength, donor site pain and muscle strains after surgery, whilst the impact of additional Gracilis tendon harvest remains unclear. Traditional graft fixation methods using a tibial interference screw generally require both Semitendinosus and Gracilis tendons, but newer techniques allow for a short single tendon (Semitendinosus) graft and suspensory fixation on the tibia with the theoretical benefit of minimising hamstring donor site morbidity and possibly graft laxity. This study aims to prospectively compare outcomes between single-tendon (ST) and dual-tendon (DT) hamstring grafts for ACL reconstruction. **Methods:** This prospective randomized controlled trial (RCT) allocated 138 patients to ACLR with one of the two hamstring tendon graft options (ST=71, DT=67) followed by a structured rehabilitation program. All anaesthetic and surgical techniques were uniform between the groups aside from graft technique and tibial fixation-Adjustable 'UltraButton' for ST and 'BioRCL' interference screw for DT (both Smith & Nephew, Boston USA). Patients were assessed pre-surgery and at 3, 6, and 12 months, with a range of patient-reported outcome measures (PROMs) including the IKDC, Lysholm and Cincinnati Knee Scores, as well as a validated donor-site morbidity score. Graft laxity, peak isokinetic knee extensor and flexor strength and a 6-hop performance battery were assessed. Limb Symmetry Indices (LSIs) were calculated. **Results:** There have been no patients lost to follow up at 1 year. No significant group differences ($p > 0.05$) were observed in patient demographics (age, sex, height and body mass) or surgical characteristics (concomitant meniscal repairs and tenodeses). While graft diameter was ≥ 8 mm in all cases, mean graft diameter was significantly smaller in the ST group compared to the DT group (mean difference [MD], -0.67mm; 95% CI, -0.91 to -0.43; $p < 0.001$). All PROMs significantly improved ($p < 0.0001$) and were largely similar between groups ($p > 0.05$). No group differences were observed for knee laxity or LSIs for hop measures ($p > 0.05$). While the ST group demonstrated a greater peak knee flexor torque LSI at 6 months ($p = 0.020$), no other strength-based LSI group differences were observed. Significant differences however were seen regarding graft site morbidity, with significantly lower morbidity reported in the ST group at 6 ($p = 0.002$) that persisted at 12 months ($p = 0.017$). **Conclusion:** To the best of our knowledge this study provides the first demonstration in a prospective, randomised trial of significantly reduced donor site morbidity using single tendon hamstring ACL grafts with adjustable suspensory tibial fixation. This results in a smaller mean graft diameter, but there is no compromise on other post-operative outcomes including PROMs, knee laxity and functional assessments. Ongoing longer-term clinical and MRI follow-up is continuing to further validate these results.

Category: Knee - ACL**Published Post Operative ACL-R Rehabilitation Programs are not Clinically Reproducible: A Scoping Review**

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Summary:

Post-operative rehabilitation programs in published peer-reviewed articles describing ACL-R for patients with a primary ACL rupture are largely clinically irreproducible.